

MECHANICAL DESIGN PORTFOLIO

MUHAMMAD ABDULLAH

Mechanical Design Engineer

Mechanical design, CAD modelling, CNC manufacturing, fabrication, sheet metal work, and product development.

CURRENT ROLE

Mechanical Design Engineer

LOCATION

Rawalpindi, Pakistan

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PROFILE / ABOUT

Design. Validate. Manufacture.

Mechanical design engineer with an emphasis on mechanisms, sheet metal design and large assemblies in SolidWorks. Experienced in CAD modelling, FEA (Ansys), engineering drawings, fabrication, machining preparation and prototyping.

Producing drawings that are precise for engineering review and practical for fabrication.

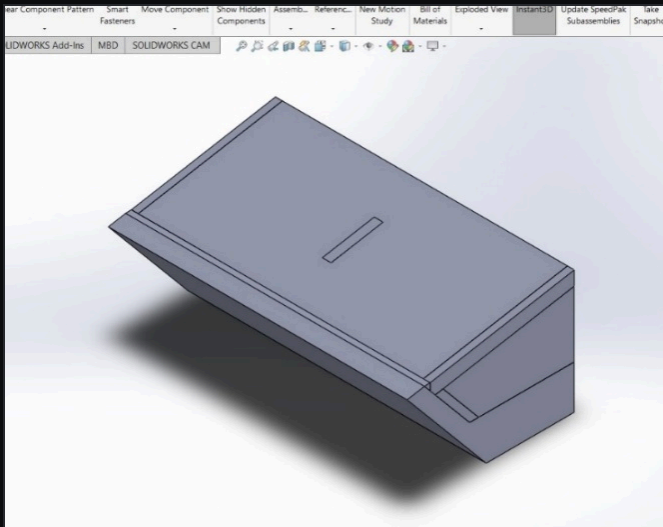
CAPABILITY / SKILLS

Capability Stack

SolidWorks	Thermal Analysis	AutoCAD	ANSYS Mechanical
CNC Programming	G/M Code	Structural Analysis	Fabrication
Prototype Development	Engineering Drawings	Mechanical Assemblies	Project Management

FINAL YEAR PROJECT

Assessment of Welding Quality on RHA Plates



Complete Al-Zarrar hull tank model developed in SolidWorks, followed by ANSYS simulation and welding analysis. Studied the effect of a 3-pass welding approach with different welding speeds and currents to optimize structural performance and reduce residual stresses and deformation.

13%

internal stress
reduction

<1 mm

deformation

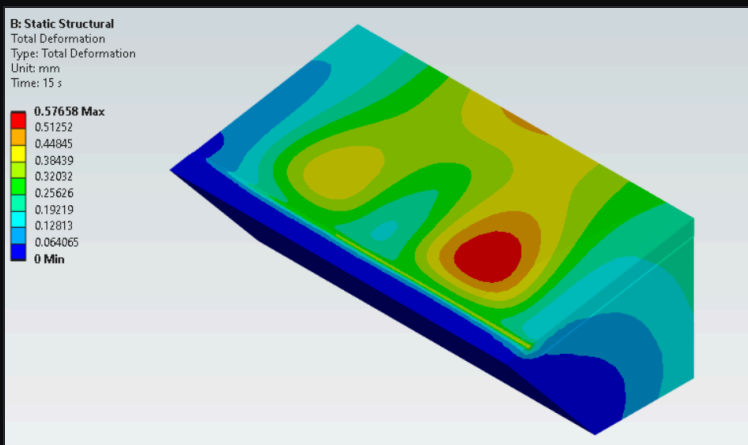
3-PASS

welding
approach

FINAL YEAR PROJECT | DESIGN VALIDATION & ANALYSIS

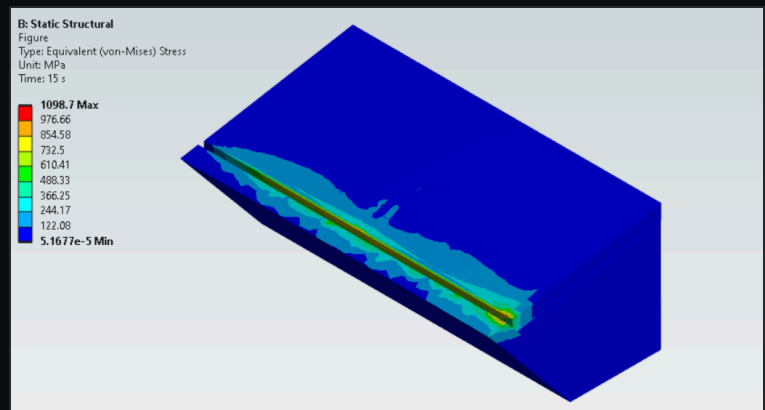
Structural and Thermal Analysis

This FYP focuses on the structural and thermal analysis of the Al-Zarrar hull tank model. The design was validated under operational loads and thermal conditions using ANSYS, ensuring the structural integrity, thermal performance and reliability of the hull component.



0.57 mm

Maximum deformation

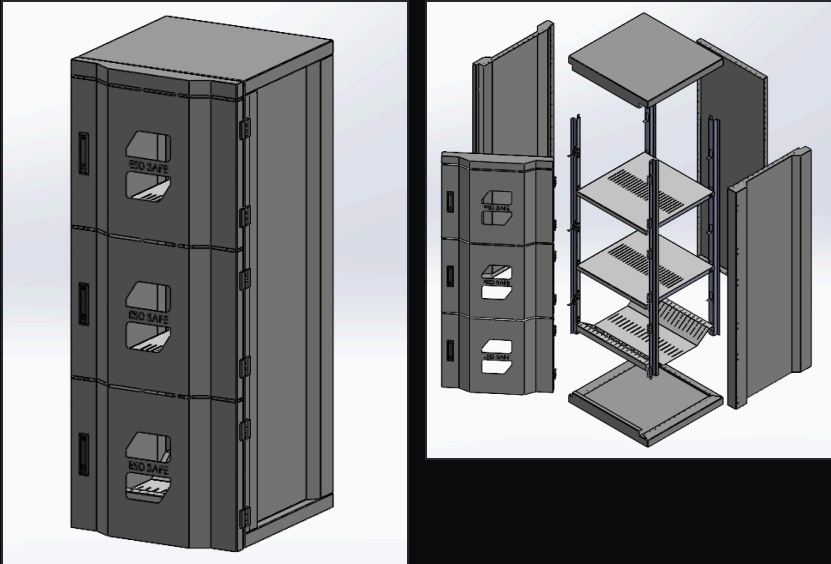


1098 MPa

Maximum von-Mises stress

Thermal

Temperature distribution & heat
effects



PROFESSIONAL SOLIDWORKS PROJECT

Advanced ESD Storage Cabinet Design at Tech Valley

ESD Protection

Ensures safe handling and storage of ESD-sensitive components with proper grounding and conductive materials.

Modular Design

Flexible and organized shelving system for efficient space utilization and component management.

Durable Construction

Robust design with high-quality materials for long service life in industrial environments.

Practical & Manufacturable

Designed in SolidWorks with attention to assembly, fabrication and real-world manufacturing.

PROJECT TYPE

Professional SolidWorks Design

KEY FOCUS

ESD Safety, Mechanical Design, Space Optimization

SOFTWARE USED

SolidWorks

Advanced MTS Systems Design at Tech Valley Islamabad

Mechanical Design

Detailed mechanical design of the MTS system with optimized structure and component integration.

3D Modeling & Assemblies

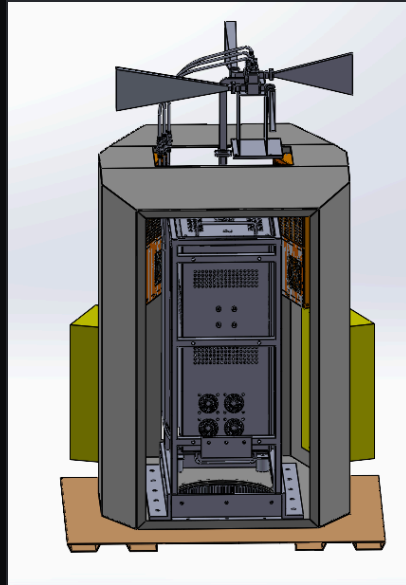
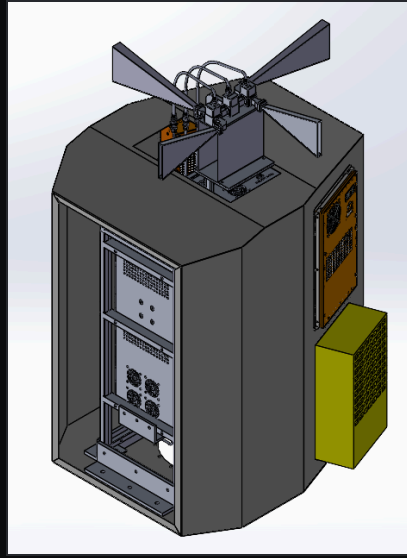
Accurate 3D CAD modeling and assembly design for full system visualization and validation.

Robust Structure

Sturdy and reliable design to ensure stability, durability and long-term performance.

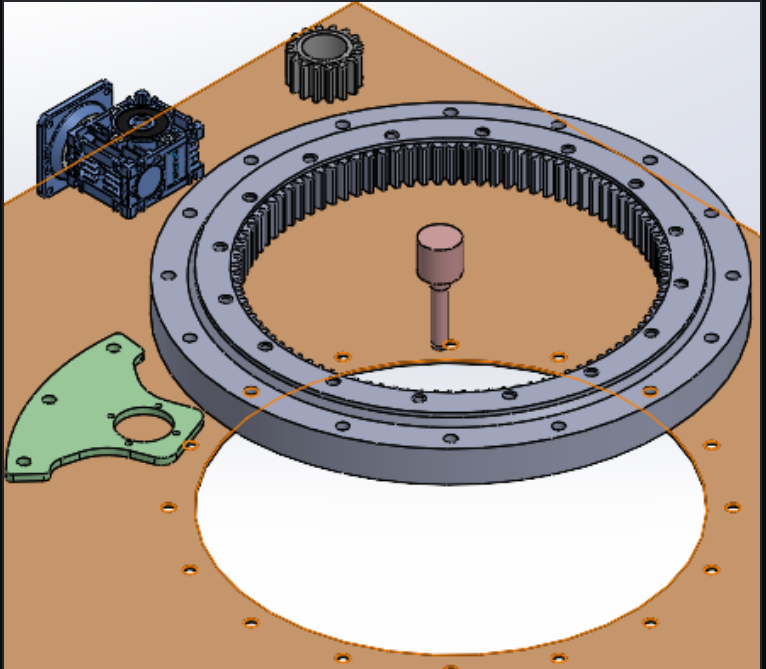
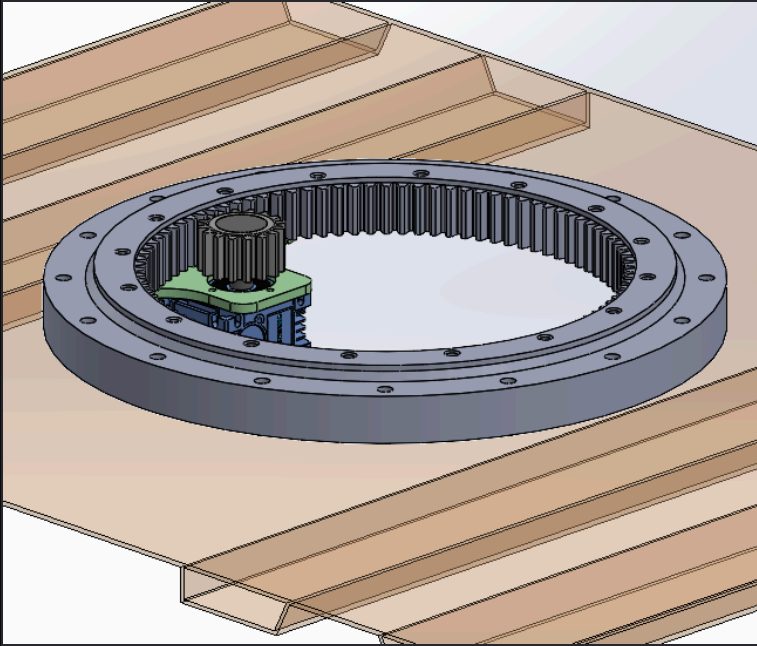
Integrated Systems

Seamless integration of mechanical, electrical and support components within the MTS unit.



Advanced Gear Mechanism Design at Tech Valley Islamabad

Designed and developed a mechanical gear mechanism in SolidWorks for rotational movement of a slew bearing. The worm gear, shown in blue, is mounted to the sheet-metal support through a shaft and drives the slew bearing for controlled rotation.



PROJECT TYPE

Mechanical Mechanism Design

KEY FOCUS

Worm Gear & Slew Bearing

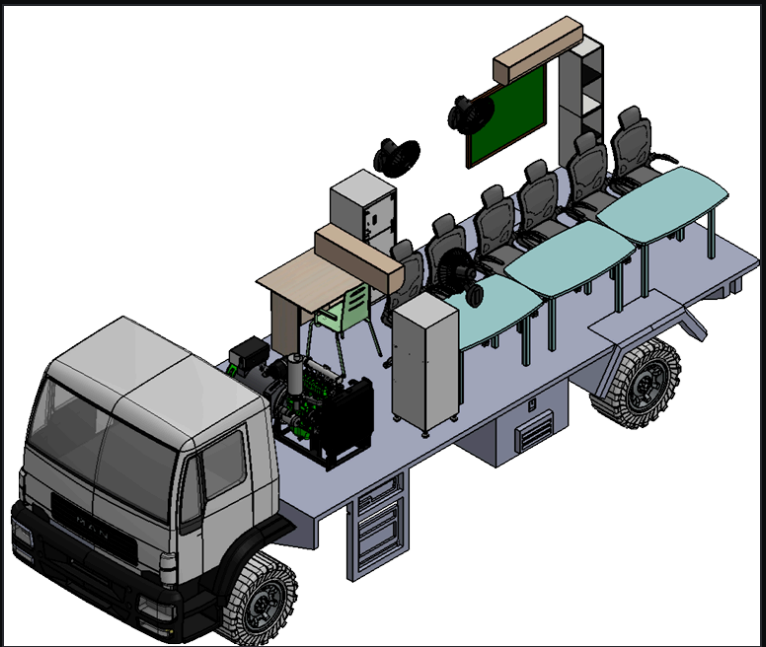
SOFTWARE USED

SolidWorks

PROFESSIONAL PROJECT | MECHANICAL DESIGN

Mobile Aerospace Workshop Cabin Design at Tech Valley Islamabad

Developed a 3D CAD layout of a mobile aerospace workshop cabin mounted on a heavy-duty MAN truck chassis for field support and maintenance operations, integrating the cabin structure and internal workstations with the vehicle platform.

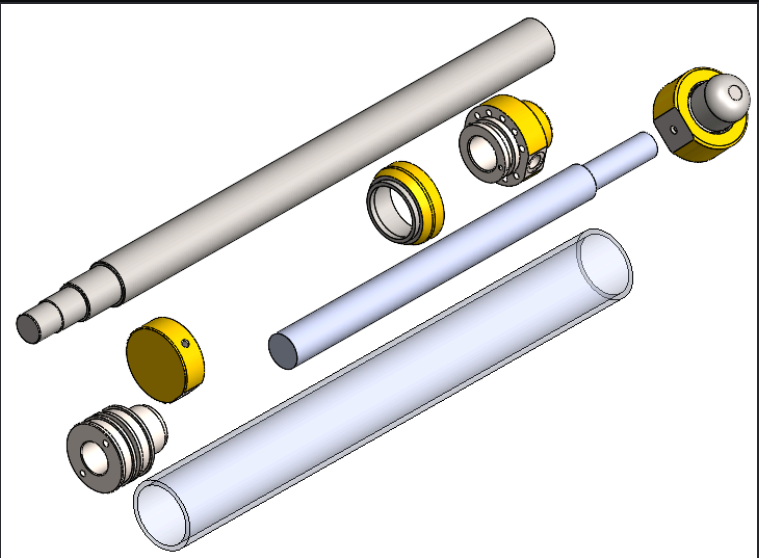
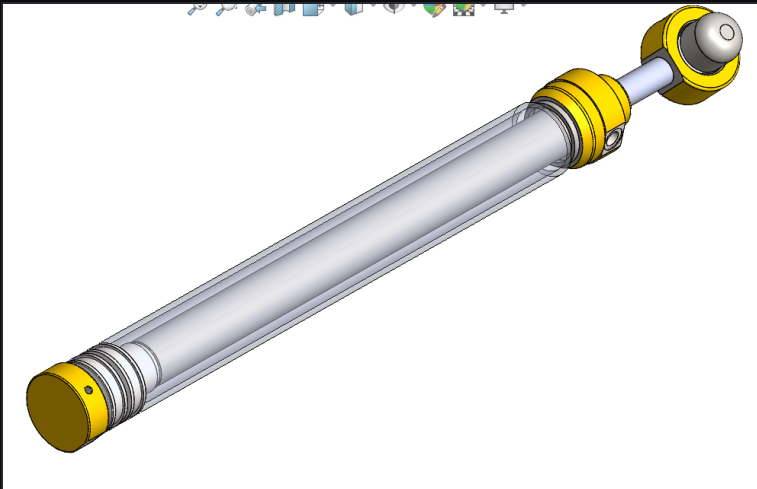


PROJECT TYPE	KEY FOCUS	SOFTWARE USED
3D CAD Modelling & Vehicle Integration	Mobile Workspace Layout	SolidWorks

PROFESSIONAL PROJECT | MECHANICAL DESIGN

Hydraulic Cylinder Assembly Design & Sealing System

Designed a complete hydraulic cylinder assembly in SolidWorks, including the piston rod, cylinder tube, gland/bushing, seal kit, and rod-eye and clevis end fittings. The exploded view details the seal arrangement and end-fitting stack-up for assembly and maintenance reference.



PROJECT TYPE	KEY FOCUS	SOFTWARE USED
Mechanical Design & Assembly	Sealing System & Rod-End Fittings	SolidWorks

Professional Background

15 Jul 2026 — Current

Mechanical Design Engineer

Tech Valley, Islamabad

Designed complex systems such as the ESD Storage Cabinet and MTS System in SolidWorks, and also worked on their manufacturing.

Aug 2025 — Jul 2026

Final Year Project — ARDIC

Heavy Industry Taxila (HIT)

Jun 2025 — Jul 2025

Intern

DESCOM (HIT)

Jun 2024 — Jul 2024

Intern

Heavy Mechanical Complex (HMC)

EDUCATION & CERTIFICATIONS

Education

2022–2026

BS Mechanical Engineering

HITEC University Taxila | CGPA 3.5 / 4.0

2020–2022

F.Sc Pre-Engineering

The Scholars Science College, Wah Cantt |
80%

Certification

FEA Using ANSYS

Finite Element Analysis training and application.

AVAILABLE FOR MECHANICAL DESIGN
OPPORTUNITIES

Let's turn a
complex
requirement into a
working design.

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